

# Bayesian Optimized Spectral Filters Coupled With Ternary ECOC for Single-Trial EEG Classification

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**Abstract**—Motivated by the promising emergence of brain–computer interfaces (BCIs) within assistive/rehabilitative systems for therapeutic applications, this paper proposes a novel Bayesian framework that simultaneously optimizes a number of subject-specific filter banks and spatial filters. Optimized double-band spectro-spatial filters are derived based on common spatial patterns coupled with the error-correcting output coding (ECOC) classifiers. The proposed framework constructs optimized subject-specific spectral filters in an intuitive fashion resulting in creation of significantly discriminant features, which is a crucial requirement for any EEG-based BCI system. Through incorporation of the ECOC approach, the classification problem is then modeled as communication over a noisy channel where the misclassification error is corrected by error correction techniques borrowed from an information theory. This paper also proposes a modified version of the ECOC adopted to EEG classification problems by deploying ternary class codewords to increase the Hamming distance between the codewords and introduce more robustness to misclassification error. The proposed framework is evaluated over two different datasets from the BCI Competition (i.e., BCIC-IV<sub>2a</sub> and BCIC-IV<sub>2b</sub>). The results indicate that the proposed approach outperforms its counterparts and validate the essential role of optimized spectral filters on the overall classification accuracy.

**Index Terms**—Brain-computer interfaces (BCI), electroencephalography (EEG), spatio-spectral filter optimization, biomedical signal processing, motor imagery, discriminative feature extraction.

## I. INTRODUCTION

**H**UMAN brain, thanks to its vast and complex network of neurons, constitutes the most powerful signal processing system ever known to us in the sense that it can simultaneously analyze and fuse several streaming signals from different modalities in an adaptive and real-time fashion. Such intriguing capabilities of the brain have motivated extensive amount of research on developing Brain-computer Interfaces

(BCI) [1]–[3] to establish direct communication links for interaction with physical world via brain signals. The BCI systems are considered as the main building block of the human-in-the-loop, cyber-physical systems [4], [5], and are becoming a key component for assistive/rehabilitative systems [6]–[8], and neuro-prosthesis control [9]. The plasticity properties of the brain [10] have opened up various possibilities for the BCI systems in therapeutic applications and have significantly enhanced the effectiveness of rehabilitation therapies. Nowadays, rehabilitation-based BCIs [11] are gaining more attention in different areas such as neuro-feedback (NFB), therapy for autism spectrum disorder (ASD), attention deficit hyperactivity disorder (ADHD), schizophrenia, and motor rehabilitation for post-stroke patients to name a few. Despite the promising vision that full integration of BCIs in rehabilitation systems presents, the research in this area is still in its early stages as the performance of the aforementioned artificial systems rarely matches that of the brain. This calls for an urgent quest to further improve the performance of existing BCI systems by developing advanced and improved processing solutions, which is the focus of this paper.

A BCI system, typically, consists of different components including the following two main building blocks: (i) A signal recording or brain imaging modality such as Electroencephalogram (EEG), Near Infra-red (NIR), Electro-corticogram (ECoG), or Magnetic Resonance Imaging (MRI) to monitor brain activities, and; (ii) A signal processing module, which is utilized to extract meaningful information from the recordings of the selected imaging modality. Having the affordability, portability, and high temporal resolution taken into account as desired features of a brain imaging modality, EEG is considered as the prime choice and is widely deployed in various practical BCI system. Processing of EEG signals generally includes two major tasks, i.e., “*Feature Generation*,” and “*Feature Translation*”. While the former is mainly concerned with the pre-processing and filtering of the EEG signals to extract informative features capable of describing the intended underlying phenomena, the latter deals with utilization of the extracted features for classification and discrimination of intended tasks (e.g., actions, mental states, and/or emotions). Next, we briefly review these two categories to better position the paper’s contribution within the existing literature.

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### A. Feature Generation

When it comes to extracting features for motor-related studies based on EEG signals, the sensorimotor activities [12] are believed to be the most informative representatives. Sensorimotor activities occur mainly in the frontal and parietal cortices of the brain within  $\mu$  (8-13 Hz) and  $\beta$  (13-30 Hz) bands. In case of a voluntary movement, a slow drop in the power of the recorded signals within the  $\mu$  band is observed (referred to as Event-Related Desynchronization (ERD)), and once the action is done, a fast increase in the power of the signals is observed within the  $\beta$  band (referred to as the Event-Related synchronization (ERS)). The ERD and ERS are also known to be correlated with motor imagery (MI) tasks [13]. Among various processing techniques proposed for spatial filtering and extracting sensorimotor activities, the Common Spatial Patterns (CSP) [14] is considered as one of the successful techniques [15]. The CSP, which is basically developed for classification of two tasks, derives a linear transformation (referred to as a spatial filter) that extracts discriminative features based on the ERD and ERS events. In other words, CSP aims at maximizing the variance of one MI task while minimizing the variance of the other task [16]. The superiority of the CSP approach in discriminating MI tasks has motivated several recent research works [17]–[23] extending it to further enhance the classification performance. Generally speaking, extensions to the CSP technique can be classified into two major categories: (i) Filter bank-based solutions such as filter bank common spatial patterns (FBCSP) [18], regularized common spatial patterns (RCSP) [19], [20], and separable common spectro-spatial patterns (SCSSP) [21]), which decompose the EEG signals into several frequency bands, and; (ii) The second group such as [23], focus on deriving a single optimized frequency band for each subject to boost the CSP performance. Reference [24] employed a filter bank approach consisting of 3 spectral filters with fixed ranges, hence referred to as a Ternary approach.

### B. Feature Translation

In case of dealing with classification of two classes of data, binary classifiers such as support vector machines (SVM) or linear discriminant analysis (LDA) are used. These methods are inherently designed for two-class problems and have shown superior performance in a different number of applications. On the other hand, when it comes to multi-class problems, these techniques are not directly applicable and new classifiers for multi-class problems should be deployed. The following two main approaches are used to address the multi-class problems: (i) Developing new classifiers, which are intrinsically designed for multi-class problems such as k-nearest neighbors (k-NN), and; (ii) Modifying the existing binary classifiers to fit multi-class problems.

### C. Contribution

The filter bank methods have demonstrated promising results over different testing scenarios, however, a significant portion of the extracted features are not utilized in the

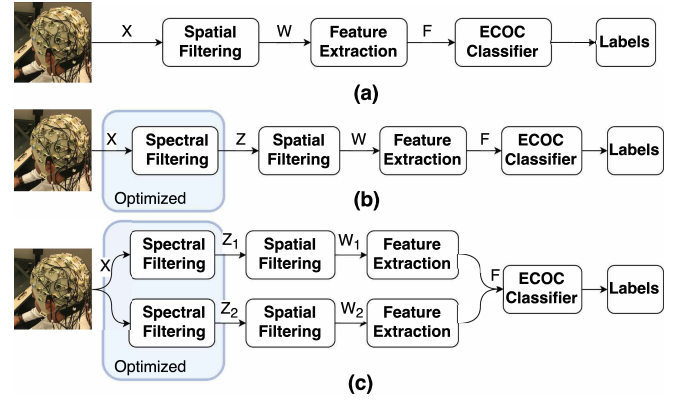


Fig. 1. The ECCSP framework: (a) The ECCSP. (b) The ECCSP-TB. (c) The ECCSP-TB2B.

classification step as some of the spectral ranges do not provide informative input for the classification, specially when fixed and general frequency bands are employed. On the other hand and to the best of our knowledge, the methods utilizing the optimized spectral filters, have only inspected the optimization of a single frequency band. We bridge the two worlds by devising an optimization framework for filter bank optimization. In addition, for multiclass classification problems, the efficacy of the classifier plays an important role in the overall performance of the processing pipeline and to this end, we introduce a new classification paradigm based on ternary numerical system shown to boost the overall performance.

To address the former foregoing drawback, we aim to couple filter banks with optimized techniques to simultaneously take advantage of both solutions. In this regard, we propose a Bayesian framework to optimize the limits of two bandpass filters. To address the latter drawback, one can extend conventional binary classifiers (e.g., SVM and LDA) to multi-class settings (like multi-class SVM), or adapt current binary classifiers to become applicable to multi-class scenarios, e.g., popular techniques in this category are “one vs. one” and “one vs. all”. In this paper, we propose an alternative solution, i.e., to incorporate the error correcting output coding (ECOC) classifiers, which splits the classification problem into a number of binary ones. The ECOC [25] has been successfully used in other application domains, in particular, it has been effectively applied for text classification problems [26]. Despite the success of the ECOC in other fields, it has not yet been used for EEG classification in the BCI systems, which is one scope of this paper. In other words, we propose to deploy the ECOC and a modified version, we refer to as the *Ternary-ECOC*, as the classification component of the feature generation methods. In summary, the paper proposes a novel processing framework for MI classifications tasks for BCI, referred to as the ECCSP. The overall theme of the proposed framework is demonstrated in Fig. 1 and is briefly outlined below:

1. **The ECCSP**, which is a combination of ECOC with CSP, where we perform spatial filtering via CSP technique and introduce ECOC as the classifier (Fig. 1(a)).

2. **The ECCSP-TB:** Improves each of the two underlying components of the ECCSP as follows: (i) We modify the ECOC algorithm and use its Ternary version, and; (ii) We incorporate a Bayesian framework to optimize the spatial-spectral filters prior to extracting the CSP features. The ECCSP-TB includes an optimization framework to obtain subject-specific spectral filters to enhance the classification accuracy, and updates the ECOC classifier to the proposed ternary version to further improve the classification accuracy (Fig. 1(b)).

3. **The ECCSP-TB2B:** Further improves the ECCSP framework by using two optimized frequency bands for the spectral filtering instead of the one used in the ECCSP-TB version. The ECCSP-TB2B version improves the spectral filtering stage of the ECCSP-TB framework to extract subject and frequency specific features leading to higher accuracy (Fig. 1(c)).

In short, the paper makes the following key contributions: (i) Through the proposed ECCSP framework, we demonstrate the applicability of the ECOC classifiers for EEG studies and introduce a modified version of the ECOC which boosts the performance of the overall system; (ii) The proposed ECCSP framework integrates the idea of utilizing filter banks to extract more informative features with that of deriving one subject-specific and optimized spectral filter to analyze the MI signals, and; (iii) Finally, we illustrate the effect and importance of utilization of the frequency specific features. By optimizing the spectral filter in the processing pipeline while keeping the feature extraction in its purest form, the performance is drastically improved. The rest of the paper is organized as follows: The ECCSP framework and its variants, ECCSP-TB and ECCSP-TB2B, are developed in Sections II, III and IV, respectively. Simulation results are provided in Section V. Finally, Section VI concludes the paper.

## II. THE ECCSP FRAMEWORK

In this section, we present the main building blocks of the proposed ECCSP framework, which is developed by coupling the ECOC classification approach with the CSP feature extraction methodology. Later in Sections III and IV, we introduce different extended/modified variants of the ECCSP framework in terms of the ECCSP-TB and ECCSP-TB2B versions. We note that, Reference [31] initially proposed utilization of ECOC classifiers for EEG classification, which constitutes the bases for the developments in this section. We would like to note that the TECOC approach developed in the next section and its incorporation for the EEG processing within the proposed processing pipelines in Sections III and IV are the main contributions of our work. In this paper, our focus is on supervised learning (for classifying MI tasks) from  $N_{\text{Trial}}$  number of available training EEG epochs (trials) denoted by  $\mathbf{X}_i \in \mathbb{R}^{N_{\text{ch}} \times N_t}$ , for  $(1 \leq i \leq N_{\text{Trial}})$ . Trial  $\mathbf{X}_i$  consists of  $N_{\text{ch}}$  number of EEG channel recordings, where  $N_t$  number of time samples are available per each channel. In other words, the processing is performed based on a training dataset denoted by  $\{(\mathbf{X}_i, \Omega_i)\}$ , where  $\Omega_i$  is the correct label of the  $i^{\text{th}}$  epoch. A pre-processing step, including bandpass filtering (range of 0.5 – 100Hz used by the BCI competition [28])

and notch filtering, is typically applied. Pre-processing is to remove high-frequency interferences, as well as the power line interference, to narrow down the frequency range of the EEG signals.

### A. ECOC Classifiers

In ECOC classifiers, the multiclass classification problem is decomposed into a number of binary classification problems and the whole process is modeled as communication of a binary codeword over a noisy channel. In terms of a communication system, the received codeword is noise contaminated and possibly different from the originally transmitted one. In the classification problem, each bit of the codeword is derived by solving each of the binary classifiers and the noise in the communication channel is actually the misclassification error of each classifier. If proper binary codewords, in terms of their distance from each other, are assigned to different classes, then it is possible to handle a few bits of error and reconstruct the original codeword from the noisy one. In fact, if the codewords are separated with the Hamming distance of  $d$ , then it is possible to handle  $\frac{d-1}{2}$  bit errors. For a multiclass problem with  $N_c$  classes, a binary coding matrix  $\mathbf{C}$  with  $N_c$  rows each with  $N_{\text{Bits}}$  bits length needs to be created. Each row of the coding matrix  $\mathbf{C}$  ( $\mathbf{C}_n$ ;  $1 \leq n \leq N_c$ ) represents the unique codeword associated with class  $n$ . Matrix  $\mathbf{C}$  defines the number of binary classifiers to be trained ( $\mathbf{y}^{(j)}$ ;  $1 \leq j \leq N_{\text{Bits}}$ ) and each column of  $\mathbf{C}$  represents the super-set shaping strategy for the each binary classifier. In this setting, the binary classifier  $\mathbf{y}^{(j)}$  associated with the  $j^{\text{th}}$  column of the coding matrix  $\mathbf{C}$  aims at distinguishing the trials which are labeled with 0 from the trials labeled with 1. This results in two super-sets for each classifier  $\mathbf{y}^{(j)}$  denoted by  $\mathbf{S}^{(j,0)}$  and  $\mathbf{S}^{(j,1)}$ , with their corresponding features which are extracted to be fed into the classifier. Training an ECOC, therefore, comprises of learning a set  $\mathbf{A} = \{\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(N_{\text{Bits}})}\}$  of independent binary classifiers.

To properly form the codewords with enough separability in terms of Hamming distance, for  $3 \geq N_c \geq 7$ ,  $N_{\text{Bits}} = 2^{N_c-1} - 1$  classifiers are required [30]. For  $8 \geq N_c \geq 11$ , selection of a proper subset of columns from the exhaustive code by an optimization technique is suggested. For  $11 \geq N_c$ , random code generation with hill-climbing procedure is proposed. As in this work we are dealing with a 4 class problem,  $\mathbf{C}$  with 4 rows and  $N_{\text{Bits}} = 2^{4-1} - 1 = 7$  columns need to be formed according to the following steps:

- (i) The first row of the coding matrix consists of all ones.
- (ii) Row  $2 \leq i \leq N_c$  is constructed by concatenation of alternating runs of  $2^{N_c-i}$  zeros and ones.

In this setting, the binary classifier  $\mathbf{y}^{(j)}$  associated with the  $j^{\text{th}}$  column of the coding matrix  $\mathbf{C}$  aims at distinguishing the trials which are labeled with 0 from the ones labeled with 1. This results in two super-sets for each classifier  $\mathbf{y}^{(j)}$  denoted by  $\mathbf{S}^{(j,0)}$  and  $\mathbf{S}^{(j,1)}$ , with their corresponding features which are extracted to be fed into the classifier. Training an ECOC classifier, therefore, comprises of learning a set  $\mathbf{A} = \{\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(N_{\text{Bits}})}\}$  of independent binary classifiers. An illustrative example is shown in Fig. 2 for classifying EEG epochs in to  $N_c = 4$  categories. Fig. 3 depicts all the



| Classes       | Codewords |   |   |   |   |   |   |
|---------------|-----------|---|---|---|---|---|---|
| Tongue MI     | 1         | 1 | 1 | 1 | 1 | 1 | 1 |
| Foot MI       | 0         | 0 | 0 | 0 | 1 | 1 | 1 |
| Left Hand MI  | 0         | 0 | 1 | 1 | 0 | 0 | 1 |
| Right Hand MI | 0         | 1 | 0 | 1 | 0 | 1 | 0 |

6-th Classifier

Class 2

Fig. 2. Generated ternary codewords for classifying 4 classes of data via Ternary-Error Correcting Output Coding (TECOC) approach.

| Classes       | Codewords |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---------------|-----------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Tongue MI     | 1         | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Foot MI       | 1         | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| Left Hand MI  | 1         | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| Right Hand MI | 1         | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
|               | X         | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | X |

Valid Codeword      Valid Codeword

Fig. 3. All possible 4-bit binary codewords for a 4 class problem. The columns denoted by "X" are not valid. Either of the two groups denoted by numbers from 1-7 could be employed as they are complement of each other.

#### Algorithm 1 ECOC Based EEG Classification

**Input:**  $\{(X_i, \Omega_i)\}_{i=1}^{N_{\text{Trial}}}$ , and;  $N_{\text{Bits}}$ .  
**Output:**  $N_{\text{Bits}}$  trained binary classifiers  $\{\gamma^{(1)}, \dots, \gamma^{(N_{\text{Bits}})}\}$ .  
S1. *Codebook Generation:* Form  $(N_c \times N_{\text{Bits}})$  coding matrix  $\mathcal{C}$ .  
S2. *Classifier Design:*  
1: **for**  $j \in \{1, \dots, N_{\text{Bits}}\}$  **do**  
2: Super-sets Construction:  $\mathcal{S}^{(j,0)} = \{X_i; \mathcal{C}_{i,j} = 0\}$  and  $\mathcal{S}^{(j,1)} = \{X_i; \mathcal{C}_{i,j} = 1\}$ .  
3: Binary Classifier Construction: Train classifier  $\gamma^{(j)}$  discriminate between  $\mathcal{S}^{(j,0)}$  and  $\mathcal{S}^{(j,1)}$ .  
4: **end for**

associated 16 possible codewords. Hypothesizing an unlabeled EEG epoch  $X_i$  is performed based on the set  $\Lambda$  of learned classifiers by first evaluating each classifier based on  $X_i$  resulting in the generation of the following binary-vector

$$\Lambda(X_i) = \{\gamma^{(1)}(X_i), \dots, \gamma^{(N_{\text{Bits}})}(X_i)\}. \quad (1)$$

Epoch  $X_i$  is assigned to a given class if the resulting bit-vector computed from Eq. (1) is similar to the codeword representing that class. In the likely case that the resulting bit-vector is not the same as any of the rows of the codebook  $\mathcal{C}$ , trial  $X_i$  will be assigned to the closest codeword as follows

$$\Phi(X_i) = \underset{m}{\operatorname{argmin}} \Delta(\mathcal{C}_m, \Lambda(X_i)), \quad (2)$$

where  $\Delta(a, b)$  is the number of bits in which vectors  $a$  and  $b$  differ. More specifically, first the Hamming distance between all the available codewords  $\mathcal{C}_i$  and the computed bit-vector  $\Lambda(X_i)$  given by Eq. (1) is computed. Then epoch  $X_i$  is assigned to class  $n$  which has the minimum distance among the other codewords. Before presenting the ECCSP, below we present an example to better clarify the ECOC technique.

#### B. The ECOC-Based CSP Spatial Filtering

The proposed ECCSP framework couples ECOC classification approach presented in Section II.B with CSP approach [17]. The latter forms a linear transformation matrix for projecting the EEG epochs into a lower-dimensional spatial subspace, which is then used by the former to construct  $N_{\text{Bits}}$  classifiers to learn the designed codebook  $\mathcal{C}$ . Essentially, the CSP approach simultaneously diagonalizes the normalized spatial covariance matrix of EEG epochs associated to the two super-sets to forms projection matrix that maximizes the variance of the two super-sets used by each of the  $N_{\text{Bits}}$  ECOC classifiers. Intuitively speaking, the reason that the CSP approach outperforms some well regarded analytical techniques (such as independent component analysis (ICA) or the principal component analysis (PCA)) for dimension reduction and feature extraction can be attributed to the fact that the CSP technique uses the labels of the data and handles the problem in a supervised fashion. Features required for classification/training of  $N_{\text{Bits}}$  binary classifiers are obtained by analyzing the spatially filtered signals.

Each classifier  $\gamma^{(j)}$ , for  $(1 \leq j \leq N_{\text{Bits}})$ , within the ECCSP framework, uses its classifier-specific features. More specifically, for trial  $X_i$ , at first we compute the normalized spatial covariance matrix as follow

$$\Sigma_i = \frac{1}{N_t - 1} (X_i - \mu_i)(X_i - \mu_i)^T, \quad (3)$$

where  $\mu_i$  is the column-wise mean of  $X_i$ . Typically, EEG signals are bandpassed resulting in the removal of the mean component, i.e.,  $\mu_i = \mathbf{0}$ . The full second-order statistical characteristics of the EEG epochs can then be inferred from

$$\Sigma_i = \frac{X_i X_i^T}{\operatorname{Tr}(X_i X_i^T)}. \quad (4)$$

For training classifier  $\gamma^{(j)}$ , for  $(1 \leq j \leq N_{\text{Bits}})$ , first we form its two corresponding super-sets  $\mathcal{S}^{(j,0)}$  and  $\mathcal{S}^{(j,1)}$ , and then compute  $\bar{\Sigma}^{(j,0)}$  and  $\bar{\Sigma}^{(j,1)}$  by taking the average of all spatial covariance matrices computed based on Eq. (4) from all the epochs that belong to the super-sets ( $\mathcal{S}^{(j,0)}$  and  $\mathcal{S}^{(j,1)}$ ), respectively. This process is explained in Algorithm 1. To further clarify this step, let us describe it in terms of our illustrative example, here term  $\bar{\Sigma}^{(6,0)}$  is computed by taking the average of covariance matrices that belong to the super-set  $\mathcal{S}^{(0)} = \{\text{Left hand MI}\}$ . Similarly, term  $\bar{\Sigma}^{(6,1)}$  is formed by taking the average of all normalized spatial covariance matrices that belong to the second super-set which is  $\mathcal{S}^{(1)} = \{\text{Tongue MI, Foot MI, Right hand MI}\}$ . The next step is to compute the composite spatial covariance matrix  $\Sigma^{(j,c)} = \bar{\Sigma}^{(j,0)} + \bar{\Sigma}^{(j,1)}$  and compute its eigenvalue decomposition as

$$\Sigma^{(j,c)} = V^{(j,c)} \lambda^{(j,c)} [V^{(j,c)}]^T, \quad (5)$$

where matrix  $V^{(j,c)}$  is constructed based on the eigenvectors of  $\Sigma^{(j,c)}$ , and the diagonal matrix  $\lambda^{(j,c)}$  is constructed from eigenvalues of the composite matrix  $\Sigma^{(j,c)}$ . These two terms are then used within the ECCSP framework to form the following whitening transformation matrix  $P^{(j)} = \sqrt{[\lambda^{(j,c)}]^{-1}} [V^{(j,c)}]^T$ , which equalizes the variance in the

**Algorithm 2** Training in the ECCSP Framework**Input:** Training EEG Trials  $\mathbf{X}$ ; The coding matrix:  $\mathbf{C}$ .**Output:**  $N_{\text{Bits}}$  trained binary classifiers  $\{\gamma^{(1)}, \dots, \gamma^{(N_{\text{Bits}})}\}$ .

- S1. According to the coding matrix  $\mathbf{C}$ , construct the two super-sets for each classifier.
- S2. Perform spatial filtering on the recordings by means of Eq. (7).
- S3. Select the first and last  $m$  rows of the spatially filtered signals.
- S4. Extract classifier-specific features using Eq. (8).

space spanned by the composite matrix's eigenvectors resulting in the eigenvalues of  $\mathbf{P}^{(j)} \bar{\Sigma}^{(j,c)} [\mathbf{P}^{(j)}]^T$  to become one. The whitening matrix  $\mathbf{P}^{(j)}$  is then used to transform matrices  $\bar{\Sigma}^{(j,0)}$  and  $\bar{\Sigma}^{(j,1)}$  (the average covariance matrices associated with each of the two super-sets) as

$$\mathbf{S}^{(j,1)} = \mathbf{P}^{(j)} \bar{\Sigma}^{(j,1)} [\mathbf{P}^{(j)}]^T \quad \text{and} \quad \mathbf{S}^{(j,0)} = \mathbf{P}^{(j)} \bar{\Sigma}^{(j,0)} [\mathbf{P}^{(j)}]^T. \quad (6)$$

The transformation introduced in Eq. (6) results in matrices  $\mathbf{S}^{(j,0)}$  and  $\mathbf{S}^{(j,1)}$  to have common eigenvectors given by  $\mathbf{S}^{(j,1)} = \mathbf{B}^{(j)} \lambda^{(j,1)} [\mathbf{B}^{(j)}]^T$  and  $\mathbf{S}^{(j,0)} = \mathbf{B}^{(j)} \lambda^{(j,0)} [\mathbf{B}^{(j)}]^T$ , where the sum of the two eigenvalues equals an identity matrix  $\mathbf{I}$  of appropriate dimension (i.e.,  $\lambda^{(j,0)} + \lambda^{(j,1)} = \mathbf{I}$ ). This neat property results in the eigenvector associated with the largest eigenvalue of  $\bar{\mathbf{S}}^{(j,0)}$  to have the smallest eigenvalue associated with  $\bar{\mathbf{S}}^{(j,1)}$  making the eigenvectors  $\mathbf{B}^{(j)}$  appropriate for classifying the two super-sets. The whitened EEG epochs are then projected onto the eigenvectors associated with the largest and smallest eigenvalues of matrix  $\mathbf{B}^{(j)}$  to form the optimal feature vectors (in the least square sense) to discriminate the two super-sets. The projection matrix associated with classifier  $\gamma^{(j)}$  is, therefore, computed as follows  $\mathbf{W}^{(j)} = [\mathbf{P}^{(j)}]^T \mathbf{B}^{(j)}$ , which is finally used to decompose each epoch  $\mathbf{X}_i$  as

$$\mathbf{W}_i^{(j)} = [\mathbf{W}^{(j)}]^T \mathbf{X}_i. \quad (7)$$

The  $j^{\text{th}}$  classifier then constructs its feature vector  $\mathbf{f}_i^{(j)}$  from the  $i^{\text{th}}$  EEG trial  $\mathbf{X}_i$  by only using the first and last  $m$  rows of  $\mathbf{W}_i^{(j)}$  [17] as follows

$$\mathbf{f}_i^{(j)} = \log \left( \frac{\text{var}(\mathbf{W}_{i,p}^{(j)})}{\sum_{k=1}^{2m} \text{var}(\mathbf{W}_{k,p}^{(j)})} \right), \quad (8)$$

where matrix  $\mathbf{W}_{i,p}^{(j)}$  contains the first and last  $m$  rows of matrix  $\mathbf{W}_i^{(j)}$  and  $\text{var}(\cdot)$  denotes the variance operator. This completes the introduction of the proposed ECCSP framework summarized in Algorithm 2.

### III. THE ECCSP-TB

In this section, we propose a modified variant of the ECCSP, referred to as the ECCSP-TB, which deploys a Bayesian framework to optimize the spectral filters within the ECCSP

framework resulting in a set of class discriminative frequency bands and their associated spatial filters. The ECCSP-TB variant is built upon the Bayesian Spatio-Spectral Filter Optimization (BSSFO) technique proposed in [23] (reviewed below). The ECCSP-TB extends the spatial filtering of the ECCSP by utilization of the BSSFO technique and also extends the multi-class classification approach by using a ternary version of the ECOC classifier. In the following sub-sections, we present details of the ECCSP-TB variant.

#### A. Bayesian Spectra-Spatial Filter Optimization

The ECCSP-TB performs spectral filtering and spatial filtering on the EEG signals as described below.

1) *Spectral Filtering*: This step is modeled as the convolution of the input EEG epoch  $\mathbf{X}_i$  with the impulse response of a bandpass filter  $h$ , i.e.,

$$\mathbf{Z}_i = h \circledast \mathbf{X}_i, \quad (9)$$

where  $\circledast$  denotes the convolution operator, and the cutoff frequency vector  $\mathbf{B}$  specifying the bandpass filter  $h$  is defined as

$$\mathbf{B} \triangleq [b_s, b_e]^T. \quad (10)$$

In the ECCSP-TB framework, the cutoff frequencies are modeled as random variables and are optimized iteratively to increase the classification accuracy. In other words, we model the uncertainty in the cut-off frequencies of the spectral filters with a prior probability denoted by  $p(\mathbf{B})$  over random variable  $\mathbf{B}$ . The prior density  $p(\mathbf{B})$  describes relative probabilities of different states (frequency bands) in which a single-trial EEG recording is correctly discriminated. The posterior probability distribution is then computed based on each bandpass-filtered single-trial EEG recording  $\mathbf{Z}_i$ , for  $(1 \leq i \leq N_{\text{Trial}})$ , and its corresponding label denoted by  $\Omega_i$ , as

$$p(\mathbf{B}|\mathbf{Z}_i, \Omega_i) = \frac{p(\mathbf{Z}_i, \Omega_i|\mathbf{B})p(\mathbf{B})}{p(\mathbf{Z}_i, \Omega_i)}. \quad (11)$$

However, term  $p(\mathbf{Z}_i, \Omega_i|\mathbf{B})$  on the right-hand side (RHS) of Eq. (11) is too complex in nature resulting in complex  $p(\mathbf{B}|\mathbf{X}_i, \Omega_i)$ , which eliminates the possibility of direct evaluation in closed-form. Alternatively, particle-based approximation techniques [27] are utilized to address this issue. In brief, a set of  $N_p$  particles  $\{\mathbb{B}(k)\}_{k=1}^{N_p}$  generated from the prior density  $p(\mathbf{B})$  are utilized, where  $\mathbb{B}(k)$  denotes a particle representing a single filter bank. To be more specific, each particle set  $\mathbb{B}(k) = \{b_s(k), b_e(k), \pi(k)\}$  contains the characteristics of the spectral filter ( $b_s(k), b_e(k)$ ) together with its associated weight ( $\pi(k)$ ). Hence, we can rewrite the particle-based posterior distribution (Eq. (11)) as

$$p(\mathbb{B}(k)|\mathbf{Z}_i(k), \Omega_i) = \frac{p(\mathbf{Z}_i(k), \Omega_i|\mathbb{B}(k))p(\mathbb{B}(k))}{p(\mathbf{Z}_i(k), \Omega_i)}. \quad (12)$$

The posterior  $p(\mathbb{B}(k)|\mathbf{Z}_i(k), \Omega_i)$  provides all the required information regarding  $\mathbf{B}$  which can be obtained from the bandpass-filtered signal  $\mathbf{Z}_i(k)$  and its corresponding class label  $\Omega_i$ . The spectral filtering step is then followed by computing the common spatial patterns of each trial in each particle (i.e.,  $\mathbf{Z}_i(k)$ ).

2) *Spatial Filtering*: This step is performed using the CSP method (Eq. (7)). However, since particles are deployed to solve the optimization problem, we have the spatial filtering coupled with spectral filtering, i.e., Eq. (7) modifies as

$$\mathcal{W}_i^{(j)}(k) = [\mathbf{W}^{(j)}(k)]^T \mathcal{Z}_i(k). \quad (13)$$

Please note that from this stage, we add the superscript “ $(j)$ ” ( $1 \leq j \leq N_{\text{Bits}}$ ), to the equations to demonstrate that the ECOC formulations are now in effect. Eq. (13) performs the spatial filtering of the spectrally filtered signals and prepares them for the feature extraction step which is introduced in Eq. (8). After calculating the features for each trial based on the characteristics of each particle, we form a feature matrix which contains all the features for all the trials as follows

$$\mathbf{F}^{(j)}(k) = \{\mathbf{f}_{i,p}^{(j)}(k)\}_{i=1}^{N_{\text{Trial}}} \in \mathbb{R}^{2m \times N_{\text{Trial}}}. \quad (14)$$

Since the spectro-spatial filtering is a deterministic procedure and does not add any stochastic behavior to the system, we can rewrite the posterior probability in Eq. (12) as follows

$$\begin{aligned} p(\mathbb{B}(k)|\mathcal{Z}_i(k), \Omega_i) &\triangleq p(\mathbb{B}(k)|\mathbf{F}^{(j)}(k), \Omega) \\ &= \frac{p(\mathbf{F}^{(j)}(k), \Omega|\mathbb{B})p(\mathbb{B}(k))}{p(\mathbf{F}^{(j)}(k), \Omega)}. \end{aligned} \quad (15)$$

In order to compute the likelihood probability  $p(\mathbf{F}^{(j)}(k), \Omega|\mathbb{B})$  in Eq. (15), an information-theoretic approach is deployed. In this approach, the mutual information [29] is utilized to compute the likelihood function and measure the discriminative power of the features in classifying the EEG trials. In this regard, the likelihood is defined as

$$p(\mathbf{F}^{(j)}(k), \Omega|\mathbb{B}) = \exp \left[ I(\mathbf{F}^{(j)}(k); \Omega) \right], \quad (16)$$

where  $I(\cdot)$  denotes the mutual information between the features matrix and the class labels. Please note that the mutual information is actually calculated between the columns of the features matrix (i.e., each feature vector) and the class labels, therefore, it can reveal how much each feature vector is representing the class labels. The mutual information is computed based on the entropy of the features and the labels, i.e.,

$$I(\mathbf{F}^{(j)}(k); \Omega) = H(\mathbf{F}^{(j)}(k)) - H(\mathbf{F}^{(j)}(k)|\Omega), \quad (17)$$

where  $H(\cdot)$  and  $H(\cdot|\cdot)$  represent the entropy and conditional entropy, respectively [29]. Once the posterior probabilities are estimated, the weights of the particles need to be derived. The weight of each particle is computed as follows

$$\pi^{(j)}(k) = \frac{p(\mathbf{F}^{(j)}(k), \Omega|\mathbb{B}_k)}{\sum_{k=1}^{N_p} p(\mathbf{F}^{(j)}(k), \Omega|\mathbb{B}(k))}. \quad (18)$$

For each iteration of the optimization procedure, the set of particle weights are calculated according to Eq. (18). Then, the set  $\Pi^{(j)}$  is formed according to the following criteria

$$\Pi^{(j)} = \bigcup_k (\pi^{(j)}(k) > \tau). \quad (19)$$

The criteria introduced in Eq. (19) selects the particles with weights  $(\pi^{(j)}(k))$  greater than a given threshold  $\tau$ , which is a random number between 0 and 1 changing at each iteration.

If a particle is selected and is included in the set  $\Pi^{(j)}$ , its associated characteristics of the filter bank remain intact for the next iteration. However, for those particles which are not included in this set, a disturbance following a normal distribution ( $\sim \mathcal{N}(0, 1)$ ) is added to the values of the band limits, which define the characteristics of the filter banks. In the next iteration, the effect of the disturbance (changing the characteristics of the filter banks) is evaluated, and the same procedure goes on for a specific number of iterations until the particles converge to the optimum value of the band limits for the spectral filter and the particle weights. In the final step,  $N_p$  number of SVM classifiers are trained based on the features extracted from each particle.

In the evaluation phase, since we have optimized  $N_p$  number of particles,  $N_p$  different spectral filters are applied to the trials and the process is followed by applying spatial filters corresponding to each particle. Then the features of each particle are obtained and are fed into the corresponding classifier. To merge the decisions of  $N_p$  classifiers, the output of each classifier (+1 or -1) is multiplied by the weight of the particle and the final decision of the ECCSP-TB is based on the weighted summation of the outputs of the classifiers. Please note that in order to obtain a smooth classification scheme in the evaluation part, the two classes of data are labeled with +1 and -1. As stated above, this labeling comes in handy for merging the decision of the classifiers by incorporating the weight of the particles. Finally, we denote the classifiers' set for all particles as  $\{\zeta(k)\}_{k=1}^{N_p}$ , therefore, the decision rule in the ECCSP-TB is defined as

$$\lambda_{test}^{(j)} = \text{sign} \left( \sum_{k=1}^{N_p} \pi^{(j)}(k) \zeta(k)(X_{test}) \right). \quad (20)$$

This completes feature extraction of the ECCSP-TB variant, next we present the modification of the classification component in comparison to the primary ECCSP framework.

## B. Ternary ECOC

As stated previously, the ECOC technique splits the problem of multi-class classification to a number of binary sub-classification problems. The action of how to split the problem requires generating proper codewords. Up until now, deployment of binary codewords is being discussed; however, we propose to extend the ECCSP by generating codewords based on the ternary numerical system.

Intuitively speaking, we believe that the drawback of the ECOC method, introduced in section II, is that in each of the cases that we form the super-sets ( $S^{(j,0)}$  and  $S^{(j,1)}$ ), all the classes are involved and there is no chance to examine the classifier  $\mathcal{P}^{(j)}$  in absence of features of a certain class. Therefore, we propose the Ternary ECOC classifiers, where the possible labels are now 0, 1 and 2 (ternary system). In this work, we take 0 and 1 as labels to form the super-sets, and the classes with label 2, are not involved in the classifier design. By performing a similar procedure to that of the conventional ECOC technique for removing the complements and meaningless cases, and taking this fact into account that label “2” means that the class is not involved into the

**TABLE I**  
GENERATED TERNARY CODEWORDS FOR CLASSIFYING  
4 CLASSES OF DATA IN TECOC APPROACH

| Class Label   | Assigned class codeword in ternary system |
|---------------|-------------------------------------------|
| Tongue MI     | 1012101201201210121101211                 |
| Foot MI       | 0111200011122201112011120                 |
| Left hand MI  | 00000000000000000000222222                |
| Right hand MI | 0000011111111122222000002                 |

classifier training, we derive 25 particular cases. It is also worth mentioning that any 4-digit ternary number which does not consist of either 0 or 1, is defined as meaningless. Table I demonstrates the ternary codewords which are derived for classification of 4 classes of data.

If proper original codewords are assigned to the classes with enough Hamming distance, we would be able to handle, up to some extent, the misclassification error of the binary classifiers and conclude the correct class codeword. This is the main difference between the ECOC classifiers and the OVO and OVA techniques. The benefit of the proposed TECOC classification scheme over its conventional counterpart is introducing class codewords that have higher Hamming distance and, therefore, are capable of handling more misclassification errors. On the other hand, a higher number of trained classifiers and increased computational cost are the drawbacks of this technique. The decision rule for Ternary ECOC is the same as the one in Eq. (2), keeping in mind that the code words here are the ternary ones. Algorithm 3 outlines the training phase of the ECCSP-TB framework. Algorithm 4 demonstrates the work-flow in the evaluation phase.

#### IV. THE ECCSP-TB2B

In the previous section, we developed the ECCSP-TB variant, which aims at deriving the optimized spectral filters to better discriminate the EEG trials for different tasks. In the simulations results, we will show its superior performance among different competing techniques. In order to further boost the performance of the proposed ECCSP framework, in this section, we develop the ECCSP-TB2B variant. Here, the Bayesian framework developed in Section III is incorporated to optimize the characteristics of a filter bank (2 frequency bands), instead of a single spectral filter used in the ECCSP-TB variant. The motivation for this modification is to combine the benefits of the filter bank method and the optimized methods, in a way to reach higher accuracies. The reason for selecting two frequency bands is that we believe that the motory related features of the EEG signals are mainly stored in the  $\mu$  band (8 – 13Hz) and the  $\beta$  band (13 – 30Hz). Ideally speaking, it is expected that the limits of the incorporated filter bank converge such that the optimized limits maximize the useful extracted frequency contents of the aforementioned bands. In this regard, we initialize a filter bank consisting of 2 bandpass filters, and then we optimize the frequency limits based on the procedure discussed in Section III. More specifically, in the ECCSP-TB2B variant, the randomness of the cut-off frequencies for the filter bank are modeled with *a priori* probability denoted by  $p(\mathcal{B}^D)$

#### Algorithm 3 The ECCSP-TB in Training Phase

**Input:** EEG recordings  $\{X_i\}_{i=1}^{N_{\text{Trial}}}$  and their corresponding labels  $\{\Omega_i\}_{i=1}^{N_{\text{Trial}}}$

**Output:**  $N_p$  number of optimized particles  $\{\mathbb{B}(k)\}_{k=1}^{N_p}$  for each bit of ( $N_{\text{Bits}}$  in total) the ECOC classifier.

- 1: **for** The number of bits in the ECOC codeword ( $N_{\text{Bits}}$ ) **do**
- 2:   Construct the two super-sets  $\mathcal{S}^{(j,0)}$  and  $\mathcal{S}^{(j,1)}$
- 3:   Define  $N_p$  number of particles which  $\forall k : \pi(k) = \frac{1}{N_p}$ .
- 4:   Randomly initialize the filter bank band limits.
- 5:   **for** The number of iterations **do**
- 6:     **for**  $k \in$  the number of particles **do**
- 7:       Spectrally filter the signals by Eq. (9)
- 8:       Derive the CSP filters for each frequency band
- 9:       Spatially filter the signals by Eq. (7)
- 10:       Extract features by Eq. (8)
- 11:     **end for**
- 12:     Form the features matrix  $F$  by Eq. (14)
- 13:     Compute the posterior probability for Eq. (15)
- 14:     Calculate the particle weights using Eq. (18)
- 15:     **if**  $\pi(k) < \tau$  **then**
- 16:       Add normal noise to  $[b_s, b_e]$  in particle  $k$ .
- 17:     **end if**
- 18:   **end for**
- 19:   Train  $N_p$  number of SVM classifiers based on the features matrix  $F(k)$ , from the last iteration.
- 20: **end for**

#### Algorithm 4 The ECCSP-TB in Evaluation Phase

**Input:** Unlabeled EEG Trial  $X_{\text{Test}}$ ,  $N_p$  number of trained binary classifiers, optimized particles  $\{\mathbb{B}(k)\}_{k=1}^{N_p}$  and spatial filters  $W_l(k)$ .

**Output:** The predicted class label for unseen trial  $X_{\text{Test}}$ .

- 1: **for** The number of particles  $N_p$  **do**
- 2:   Spectral filtering of the trial based on Eq. (9).
- 3:   Spatial filtering of the trials according to Eq. (7).
- 4:   Extract classifier-specific features using Eq. (8).
- 5:   Obtain the prediction of each classifier for each set of features,  $\lambda$ .
- 6: **end for**
- S2. Compute the weighted summation of the predictions.
- S3. Obtain the final decision based on the sign of the weighted summation.

over random variable  $\mathcal{B}^D$ . In this scheme, three random variables are deployed to characterize the filter bank and  $\mathcal{B}^D$  is defined as

$\mathcal{B}^D \triangleq [b_s, b_m, b_e]$ , where ideally the  $\mu$  band contents are to be extracted with the  $[b_s, b_m]$  bandpass filter and the  $\beta$  band contents to be extracted with the  $[b_m, b_e]$  bandpass filter. The idea of considering a same cut-off-frequency  $b_m$  for both bandpass filters mainly relies on the typical values of the frequency bands that are reported in physiology, i.e., the  $\mu$  and the  $\beta$  bands have 13 Hz in common. Similar to the ECCSP-TB



variant, the posterior density denoted now by  $p(\mathcal{B}^D|X_i, \Omega_i)$  is constructed from single-trial EEG recording  $X_i$ , for  $(1 \leq i \leq N_{\text{Trial}})$ , and its particle-based approximation is

$$p(\mathbb{B}^D(k)|X_i, \Omega_i) = \frac{p(X_i, \Omega_i|\mathbb{B}^D(k))p(\mathbb{B}^D(k))}{p(X_i, \Omega_i)}, \quad (21)$$

where  $\mathbb{B}^D(k) = \{b_s(k), b_m(k), b_e(k), \pi(k)\}$  denotes particle  $k$ , for  $(1 \leq k \leq N_p)$ , and represents a single filter bank. The bandpass filtering of the signals is modeled as a convolution operation on the input signals performed by the system  $h_l(k)$ , for  $(l \in \{1, 2\})$ , associated with each of the two bandpass filters. Please note that  $l = 1$  determines the filter for  $[b_s, b_m]$ , and  $l = 2$  denotes the filter for  $[b_m, b_e]$ . Therefore, the filtered signal, denoted by  $\mathcal{Z}_l$ , is deterministically obtained as

$$\mathcal{Z}_{i,l}(k) = h_l(k) \circledast X_i, \quad (22)$$

The likelihood and the evidence are, therefore, become equal to  $p(\mathcal{Z}_{i,l}(k), \Omega_i|\mathbb{B}^D(k))$  and  $p(\mathcal{Z}_{i,l}(k), \Omega_i)$ , respectively. Hence, we can rewrite (21) by replacing the raw EEG signal  $X_i$  with its bandpass-filtered version  $\mathcal{Z}_{i,l}(k)$  as

$$p(\mathbb{B}^D(k)|\mathcal{Z}_{i,l}(k), \Omega_i) = \frac{p(\mathcal{Z}_{i,l}(k), \Omega_i|\mathbb{B}^D(k))p(\mathbb{B}^D(k))}{p(\mathcal{Z}_{i,l}(k), \Omega_i)}. \quad (23)$$

The spectral filtering step is then followed by computing the common spatial patterns of each trial for each frequency band in each particle (i.e.,  $\mathcal{Z}_{i,l}(k)$ ). The spatial filtering formula is rewritten as  $\mathcal{W}_{i,l}^{(j)}(k) = [\mathbf{W}_l^{(j)}(k)]^T \mathcal{Z}_{i,l}(k)$ , and the feature extraction formula is updated as

$$f_{i,l}^{(j)}(k) = \log \left( \frac{\text{var}(\mathcal{W}_{i,l,p}^{(j)}(k))}{\sum_i \text{var}(\mathcal{W}_{i,l,p}^{(j)}(k))} \right). \quad (24)$$

Now, at this stage, the features of the two parallel pipelines are merged into one single feature vector as

$$\mathbf{f}_i^{(j)}(k) = \left[ f_{i,l}^{(j)}(k)|_{l=1}, f_{i,l}^{(j)}(k)|_{l=2} \right], \quad (25)$$

and then the vectors are combined according to Eq. (14). The rest of the procedure is the same as discussed in Section III.

## V. SIMULATIONS

In this section, we present an extensive set of simulation experiments to evaluate the performance of the proposed ECCSP framework and its variants. The experiments are performed based on two well regarded MI datasets provided by BCI Competition IV, referred to as “Dataset BCIC-IV<sub>2a</sub>” and “Dataset BCIC-IV<sub>2b</sub>” [28].

**Dataset BCIC-IV<sub>2a</sub>:** This dataset consists of four MI classes of EEG measurements (Right hand MI, Left hand MI, Foot MI, and Tongue MI), obtained from 9 healthy subjects. Signals are recorded with sampling rate of 250Hz, using 22 EEG channels and 3 monopolar electrooculogram (EOG) channels (with left mastoid serving as the reference). The original EEG signal recordings are already bandpass filtered (0.5 – 100Hz) and notch filtered to remove the interference of the power line on the signals. For each subject, two sessions are recorded

(one for training purposes and the other one for evaluation). Each session consists of six segments, and each segment includes 48 trials of length 3 seconds. In total and for each subject, 288 trials for training and 288 trials for evaluation are available. In order to measure the performance of the proposed ECCSP framework and based on the recommendation from BCI competition [28], kappa coefficient  $\kappa$  is used, i.e.,  $\kappa = \frac{\text{CCR} - P_{\text{rand}}}{1 - P_{\text{rand}}}$ , where CCR represents the correct classification rate, and the value of  $P_{\text{rand}} = 0.25$ . All the algorithms are implemented only based on the 22 EEG channel recordings.

**Dataset BCIC-IV<sub>2b</sub>:** This dataset consists of the EEG recordings from 9 different right-handed subjects, with normal or corrected-to-normal vision. The subjects were asked to perform two MI tasks including “Left Hand MI” and “Right Hand MI.” From each subject, 5 sessions are recorded, 3 of them for training and the rest for evaluation. Among the training sessions, 2 of them are recorded without providing feedback to the subject, and one is recorded when the feedback is enabled. This dataset provides 6 recording channels, 3 for EEG and 3 for EOG. The EEG channels are recorded from C3, Cz, and C4 points in the 10-20 EEG recording system, similar to the previous dataset we only use EEG recordings. The signals are recorded with sampling frequency of 250Hz and bandpass filtered between 0.5 – 100Hz. A notch filter is also applied to remove the 50Hz effect of the power line on the recordings. For this dataset, the performance measure is also kappa coefficient where  $P_{\text{rand}} = 0.5$ .

### A. Parameter Selection

As stated previously in Section V.A and similar in nature to particle filter based approaches [27] there are two hyper-parameters in the proposed framework, which need to be optimized, i.e., the number of iterations and the number of particles. As simultaneous optimization of these parameters is time-consuming and computationally expensive, we have deployed a grid search approach for optimizing these two parameters in an independent fashion. The assumption of taking the parameters independent of each other is common in multi-parameter phenomena, such as naive Bayes classifiers. For each of the parameters, values from 5 to 50 with the step size of 5 are considered and the performance of the framework is evaluated through  $10 \times 10$ -fold cross-validation. The parameter selection is performed for each subject individually and subject-specific parameters are then employed for computing the final accuracies. We emphasize that only the training samples are used for the parameter selection process and test samples are not used within this process.

**1) Number of Iterations:** In this experiment we investigate the number of iterations which yield the maximum cross-validation performance. To this aim, we keep the number of particles constant equal to 30 particles. This number is reported by [23], which was the first paper proposing the utilization of a Bayesian framework for spatial-spectral filter optimization. Then the number of iterations is varied from 5 to 50 with steps of 5, and the  $10 \times 10$ -fold cross-validation of the performance is measured. The results are presented in



TABLE II

PERFORMANCE COMPARISON IN KAPPA VALUES ( $\kappa$ ) FOR DIFFERENT APPROACHES, TESTED ON  $BCIC-IV_{2a}$  FOR PREDICTION ON TEST DATA

| Subjects  | ECCSP | SCSSP<br>[21] | FBCSP<br>[21] | FBCSP<br>[18] | BSSFO<br>OVO | BSSFO<br>OVA | TSSM+SVM<br>[16] | BSSFO<br>Conventional ECOC | ECCSP-TB | ECCSP-TB2B  |
|-----------|-------|---------------|---------------|---------------|--------------|--------------|------------------|----------------------------|----------|-------------|
| Subject 1 | 0.48  | 0.62          | 0.68          | 0.68          | 0.62         | 0.31         | <b>0.77</b>      | 0.52                       | 0.60     | 0.75        |
| Subject 2 | 0.20  | 0.27          | 0.30          | 0.42          | 0.20         | 0.08         | 0.33             | 0.14                       | 0.25     | <b>0.44</b> |
| Subject 3 | 0.49  | 0.66          | 0.71          | 0.75          | 0.70         | 0.57         | 0.77             | 0.56                       | 0.64     | <b>0.80</b> |
| Subject 4 | 0.31  | 0.27          | 0.39          | 0.48          | 0.41         | 0.45         | 0.51             | <b>0.54</b>                | 0.50     | <b>0.54</b> |
| Subject 5 | 0.16  | 0.07          | 0.28          | <b>0.40</b>   | 0.09         | -0.07        | 0.35             | 0.07                       | 0.15     | 0.27        |
| Subject 6 | 0.13  | 0.26          | 0.25          | 0.27          | 0.15         | 0.08         | <b>0.36</b>      | 0.18                       | 0.24     | 0.31        |
| Subject 7 | 0.64  | 0.41          | 0.57          | 0.77          | 0.72         | 0.62         | 0.71             | 0.64                       | 0.75     | <b>0.80</b> |
| Subject 8 | 0.38  | 0.59          | 0.59          | <b>0.76</b>   | 0.58         | 0.30         | 0.72             | 0.52                       | 0.64     | 0.73        |
| Subject 9 | 0.38  | 0.66          | 0.54          | 0.61          | 0.65         | 0.55         | <b>0.83</b>      | 0.56                       | 0.69     | 0.75        |
| Average   | 0.35  | 0.42          | 0.48          | 0.57          | 0.46         | 0.32         | 0.59             | 0.41                       | 0.50     | <b>0.60</b> |

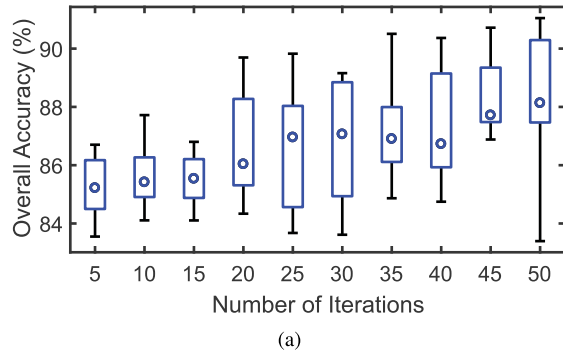


Fig. 4.  $10 \times 10$ -fold cross-validation with different number of iterations: 45 iterations are selected as the optimum one.

Fig. 4. It is observed that the number of iterations providing the best performance in terms of maximum mean and minimum variance is 45. We note that, although for 50 number of iterations higher mean performance is achieved, the variance the range of results is wider making it undesirable.

2) *Number of Particles*: Here, the number of iterations is set to 45 and we vary the number of particles within the range of 5 to 50 with steps of 5 to select the best option for the number of particles. Likewise the previous experiment,  $10 \times 10$ -fold cross-validation is deployed to measure the performance of the system. However, in contrary to the previous experiment, we optimize the number of particles separately for each individual, i.e., a subject-specific hyper-parameter selection procedure.

### B. Experimental Procedure, Results, and Discussions

Segments of 2 seconds (500 samples) from each trial are selected as the input to each framework. The segment starts 0.5 seconds after the onset cue of each trial and lasts for 2 seconds. For the spectral filters which are used in all of the algorithms, we have deployed 5<sup>th</sup> order Butterworth bandpass filter. Since the optimization procedure is time-consuming and computationally extensive, we decided to deploy a low order spectral filter but kept it the same across all the algorithms for fair comparison. For the spatial filtering of the signals, we have utilized the CSP technique in its most basic form. The reason for this selection is that our main objective is to demonstrate the effectiveness of deploying optimized spectral filter banks

on the overall performance of the system, therefore, we used basic forms of the spatial filtering and feature extraction.

After the spectral filtering step, the signals are fed to the CSP method to perform dimensionality reduction and spatial filtering. To identify the optimal number of new dimensions, for dataset  $IV_{2a}$ , 10-fold cross-validation is performed on the training samples and  $m = 2$  is selected. For dataset  $IV_{2b}$ , as only three recording channels are available,  $m = 1$  is selected for dimensionality reduction process. In the classification step, the SVM is used as the underlying binary classifier of the ECOC classifiers. It is worth mentioning that the linear SVM classifier is deployed. In the training phase and according to the procedure described in Section II, we deploy generated codewords for the 4 class classification problem which are introduced in Fig. 3 and Table I. When it comes to the evaluation phase, the extracted test trials are fed into each trained particle and are filtered according to the specific spectral characteristics of that particle. The votes of the particles are then integrated into the final decision based on Eq. (20). This whole process produces one bit out of the  $N_{\text{Bits}}$  required for the Ternary ECOC classifier to produce the final decision based on Eq. (2).

Results from eight different approaches are computed and compared for accuracy in Table II. In particular, we have implemented and compared our framework against the results of the following algorithms: (i) The FBCSP method [18], which is the winner of the BCI competition IV and achieved the highest performance for this dataset; (ii) The SCSSP method [21], which is one of the most recent works on this problem; (iii) The FBCSP results reported in Reference [21], which are computed without incorporation of the mutual information-based feature selection step; (iv) The best variant of the Tangent Space of Sub-Manifold (TSSM) approach [16], which is considered as the state-of-the-art in statistical (engineered and not data driven) solutions, and; (v)-(vi) The BSSFO [23] framework when coupled with the proposed Ternary ECOC and conventional ECOC classifiers. In addition, we have provided the simulation results for the two techniques which are served as extensions of binary classifiers for multi-class classification problems. The two techniques are called “One-Versus-One (OVO)” and “One-Versus-All (OVA),” which are extensively used for the multi-class classification problems.

TABLE III  
PERFORMANCE COMPARISON FOR DIFFERENT APPROACHES,  
TESTED ON BCIC-IV<sub>2b</sub> DATASET. PERFORMANCE  
MEASURE IS IN KAPPA ( $\kappa$ ) VALUE

| Subjects  | ECCSP       | ECCSP-TB<br>(BSSFO) | FBCSP | ECCSP-TB2B  |
|-----------|-------------|---------------------|-------|-------------|
| Subject 1 | 0.15        | 0.19                | 0.21  | <b>0.28</b> |
| Subject 2 | 0.01        | 0.10                | 0.16  | <b>0.17</b> |
| Subject 3 | <b>0.24</b> | 0.07                | -0.04 | 0.09        |
| Subject 4 | 0.37        | 0.92                | 0.61  | <b>0.96</b> |
| Subject 5 | 0.24        | 0.20                | 0.55  | <b>0.58</b> |
| Subject 6 | -0.05       | 0.45                | 0.23  | <b>0.61</b> |
| Subject 7 | 0.44        | 0.45                | 0.31  | <b>0.56</b> |
| Subject 8 | 0.74        | 0.77                | 0.18  | <b>0.80</b> |
| Subject 9 | 0.31        | 0.62                | 0.13  | <b>0.64</b> |
| Average   | 0.27        | 0.42                | 0.26  | <b>0.52</b> |

The performance of the proposed frameworks evaluated on dataset BCIC-IV<sub>2a</sub> is provided in Table II and the results for dataset BCIC-IV<sub>2b</sub> is presented in Table III. For the ECCSP-TB and ECCSP-TB2B variants, it is worth mentioning that in the first iteration, the particles are weighted equally ( $1/N_p$ ), therefore, the initial value for the particle weights is set to  $1/40$  and the band limits of the spectral filters are initialized with random numbers within the range of 4-40 Hz. Please note that for all of the techniques, the spectral filter which is used is a 5<sup>th</sup> order Butterworth bandpass filter. For the dataset BCIC-IV<sub>2b</sub>, the proposed ECCSP-TB and ECCSP-TB2B variants are employed without the TECOC classification scheme, as we are dealing with a binary classification problem. As it is observed from Table II and Table III, the proposed ECCSP-TB2B framework outperforms its variations as well as other existing frameworks. In particular and in comparison to [16], which is considered as the state-of-the-art in statistical solutions, while the overall result obtained from the proposed framework is slightly better than the best result (TSSM+SVM) reported in [16], our proposed framework outperforms this solution over 5 of the subjects and has less computational complexity during the evaluation phase in comparison to the TSSM+SVM solution.

We have included the results for the FBCSP both with the inclusion of the feature selection step and following [21], without incorporation of the feature selection procedure, as the main goal here is to investigate/illustrate the effects of employing personalized spectro-spatial filters. Therefore, to better track the effects of the optimized filters, we tried to keep the rest of the processing pipeline as simple as possible. The results in the third and forth columns of Table II are as reported in [21]. Intuitively speaking, the difference between these results with the ones reported in the original paper [28] (the fifth column of Table II) can be attributed to the feature selection technique based on mutual information of the features and the labels. The feature selection step shrinks the dimensionality of the features space from 36 to a new dimension between 4 to 8 and thus a considerable fraction of the features, which are not informative and potentially mislead the classifier are removed. The proposed framework, however, uses only two optimized and subject-specific spec-

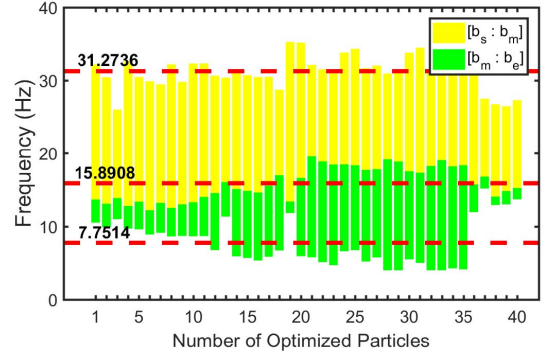


Fig. 5. Optimized frequency bands for all of the particles within 45 iterations.

tral filters which is computationally far less expensive than the FBCSP and eliminates the need for performing feature selection.

As a final experiment, we investigate the values of the frequency bands when the optimization process is finished to corroborate our intuition for deploying two frequency bands within the ECCSP-TB2B framework. Fig. 5 shows the results for subject 1 in dataset IV<sub>2b</sub>. Fig. 5 shows the converged frequency bands of the particles used to optimize the subject-specific spectral filters. As we are optimizing the cutoff frequencies of a filter bank which consists of two spectral filters, two different colors (green and yellow) are employed. Fig. 5 is introduced to support our intuition that the proposed framework more closely follows the physiological clues. Sorting the particles in descending order in terms of their associated weights reveals that the first particles have converged to the values that are more compatible with the typical values reported in physiology.

## VI. CONCLUSION

Extracting discriminative features from the EEG signals plays an important role in the performance of any EEG-based BCI system. Different factors contribute to properness of the features including the preprocessing step, the spectral filters, the spatial filters, and the feature extraction technique. In this work, we demonstrated the effect of the spectral filters on the overall performance of the system. To this aim, we deployed a Bayesian framework to optimize the characteristics of the spectral filters, to best fit them for individual's motor-cortex activity. In addition, we proposed the integration of Ternary-ECOC with the introduced optimization framework to fit the framework for multi-task MI classification problems. Finally, it is worth mentioning that recently data driven solutions (such as Convolutional Neural Networks (CNNs)) are proposed and combined with their statistical counterparts, e.g., the work in [32] combines CNN with FBCSP. It is expected to achieve better performances from such hybrid solutions as reported in [32]. An interesting direction is to use the proposed optimized spectral solution ECCSP-TB2B in combination with the CNNs, which is the focus of our ongoing research.

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